Semiconductor Physics and Devices

Lecture 4
Statistical Laws
ETE-301

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Statistical Laws

To determine the statistical behavior of particles, need to know the laws that the particles obey.

There are three distribution laws determining the distribution of particles among available energy states.

→ Maxwell–Boltzmann probability function.

→ Bose–Einstein function.

→ Fermi–Dirac probability function.

The Energy Distribution Function

Maxwell-Boltzmann (classical) $f(E) = \frac{1}{Ae^{E/kT}}$	Bose-Einstein (quantum) $f(E) = \frac{1}{Ae^{E/kT} - 1}$	Fermi-Dirac (quantum) $f(E) = \frac{1}{Ae^{E/kT} + 1}$
Identical but distinguishable particles.	Identical indistinguishable particles with integer spin.	Identical indistinguishable particles with half-integer spin (fermions).
Molecular speed distribution	Thermal radiation Specific heat	Electrons in a metal Conduction in semiconductor.

Distinguishability of Particles

The Einstein-Bose and Fermi-Dirac distributions differ from the classical Maxwell-Boltzmann distribution because the particles they describe are indistinguishable.

Particles are considered to be indistinguishable if their wave packets overlap significantly. This kind of consideration comes from the fact that all particles have characteristic wave properties according to the DeBroglie hypothesis .

Two particles can be considered to be *distinguishable* if their separation is large compared to their DeBroglie wavelength.

For example, the condition of distinguishability is met by molecules in an ideal gas under ordinary conditions.

For oxygen gas at STP, the molecules have a separation on the order of 3 nm and DeBroglie wavelengths on the order of 0.03 nm, a factor of a thousand smaller. On the other hand, two electrons in the first shell of an atom are inherently indistinguishable because of the large overlap of their wave functions.

Maxwell–Boltzmann probability function.

In this case, the particles are considered to be distinguishable by being numbered, for example, from 1 to *N*, with no limit to the number of particles allowed in each energy state. The behavior of gas molecules in a container at fairly low pressure is an example of this distribution.

The Maxwell-Boltzmann distribution is the classical distribution function for distribution of an amount of energy between identical but distinguishable particles.

$$f(E) = \frac{1}{Ae^{E/kT}}$$

the presumption of distinguishability, classical statistical physics postulates further that: There is no restriction on the number of particles which can occupy a given state. At thermal equilibrium, the distribution of particles among the available energy states will take the most probable distribution consistent with the total available energy and total number of particles. Every specific state of the system has equal probability.

Maxwell–Boltzmann function.

With increasing energy E, it is progressively less likely that any given particle will attain that energy, so more particles will be found with lower energies. It is assumed that an unlimited number of particles can occupy any energy state.

The probability that a particle will have energy E

$$f(E)_{\text{Maxwell-Boltzmann}} = \frac{1}{Ae^{E/kT}}$$

Normalization constant A

The probability for occupying a given energy state decreases exponentially with energy

Boltzmann's constant k times the absolute temperature T. The implication of this term is that for a higher temperature, it is more probable that a given particle can be found with energy E.

Bose–Einstein function.

A second distribution law is the Bose–Einstein function. The particles in this case are indistinguishable and, again, there is no limit to the number of particles permitted in each quantum state. The behavior of photons, or black body radiation, is an example of this law

The Bose-Einstein distribution describes the statistical behavior of integer spin particles (bosons). At low temperatures, bosons can behave very differently than fermions because an unlimited number of them can collect into the same energy state, a phenomenon called "condensation".

$$f(E) = \frac{1}{Ae^{E/kT} - 1}$$

Bose–Einstein function

The probability that a particle will have energy E

Describing integer spin bosons, this distribution allows an unlimited number of particles to condense into a single level.

$$f(E)_{\text{Bose-Einstein}} = \frac{1}{Ae^{E/kT} - 1}$$

For photons, A=1, so the occupation of very low energy states can increase without limit.

The exponential dependence upon energy and temperature. See the classical Boltzmann distribution for more description.

The quantum difference which arises from the fact that the particles are indistinguishable.

Fermi–Dirac probability function.

The third distribution law is the Fermi–Dirac probability function. In this case, the particles are again indistinguishable, but now only one particle is permitted in each quantum state. Electrons in a crystal obey this law. In each case, the particles are assumed to be no interacting.

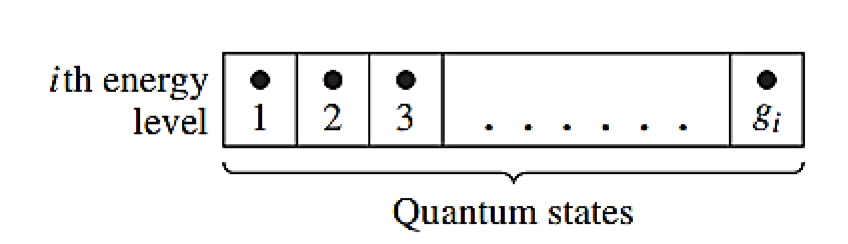

Figure shows the *i* th energy level with *gi* quantum states. A maximum of one particle is allowed in each quantum state by the Pauli exclusion principle. There are *gi* ways of choosing where to place the first particle, (*gi* -1) ways of choosing where to place the second particle, (*gi* - 2) ways of choosing where to place the third particle, and so on. Then the total number of ways of arranging *Ni* particles in the *i* th energy level (where $N_i \leq g_i$) is

$$(g_i)(g_i - 1) \cdots (g_i - (N_i - 1)) = \frac{g_i!}{(g_i - N_i)!}$$

Fermi–Dirac probability function

However, since the particles are indistinguishable, the N_i ! number of permutations that the particles have among themselves in any given arrangement do not count as separate arrangements. The interchange of any two electrons, for example, does not produce a new arrangement. Therefore, the actual number of independent ways of realizing a distribution of N_i particles in the i th level is

$$W_i = \frac{g_i!}{N_i!(g_i - N_i)!}$$

Above equation gives the number of independent ways of realizing a distribution of N_i particles in the i th level. The total number of ways of arranging (N_1 ,N_2 ,N_3 , . . . ,N_n)indistinguishable particles among n energy levels is the product of all distributions,

$$W = \prod_{i=1}^{n} \frac{g_i!}{N_i!(g_i - N_i)!}$$

The maximum *W* is found by varying *Ni* among the *Ei* levels, which varies the distribution, but at the same time, we will keep the total number of particles and total energy constant. We may write the most probable distribution function as

$$\frac{N(E)}{g(E)} = f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$$

The number density *N*(*E*) is the number of particles per unit volume per unit energy and the function *g*(*E*) is the number of quantum states per unit volume per unit energy. The function *f F*(*E*) is called the *Fermi–Dirac distribution* or probability function and gives the probability that a quantum state at the energy *E* will be occupied by an electron. The energy *EF* is called the *Fermi energy* . Another interpretation of the distribution function is that *fF* (*E*) is the ratio of filled to total quantum states at any energy *E* .

At absolute zero, fermions will fill up all available energy states below a level E_F called the Fermi energy with one (and only one) particle. They are constrained by the Pauli exclusion principle. At higher temperatures, some are elevated to levels above the Fermi level.

The probability that a particle will have energy E

$$f(E)_{\text{Fermi-Dirac}} = \frac{1}{e^{(E - E_F)/kT} + 1}$$

See the Maxwell-Boltzmann distribution for a general discussion of the exponential term.

For low temperatures, those energy states below the Fermi energy E_F have a probability of essentially 1, and those above the Fermi energy essentially zero.

The quantum difference which arises from the fact that the particles are indistinguishable.

Objective: Determine the possible number of ways of realizing a particular distribution for (a) $g_i = N_i = 10$ and (b) $g_i = 10$, $N_i = 9$.

Solution

(a) $g_i = N_i = 10$: We may note that $(g_i - N_i)! = 0! = 1$. Then, from Equation (3.77), we find

$$\frac{g_i!}{N_i!(g_i - N_i)!} = \frac{10!}{10!} = 1$$

(b) $g_i = 10$, $N_i = 9$: We may note that $(g_i - N_i)! = 1! = 1$. Then, we find

$$\frac{g_i!}{N_i!(g_i - N_i)!} = \frac{10!}{(9!)(1)} = \frac{(10)(9!)}{(9!)} = 10$$

Comment

In part (a), we have 10 particles to be arranged in 10 quantum states. There is only one possible arrangement. Each quantum state contains one particle. In part (b), we have 9 particles to be arranged in 10 quantum states. There is one empty quantum state, and there are 10 possible positions in which that empty state may occur. Thus, there are 10 possible arrangements for this case.